A construction of four-dimensional Yang–Mills theory

First sections for review; the detailed proofs are incomplete.

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Part I
The main theorem

1 The result

How can a model on a lattice give a quantum field theory on continuous space and time? We study this question for four-dimensional Yang–Mills theory, starting with Wilson's lattice model. On a finite lattice the variables are finitely many unitary matrices, and their law is an ordinary probability measure. The problem is to choose the coupling and local observables so that their correlations—the averages of their products—have a limit as the lattice becomes finer and the box becomes larger.

We construct such limits for matrix groups with simple nonabelian Lie algebra. The spacing tends to zero first, at each fixed box size. From any increasing sequence of box sizes, one can then select a subsequence along which all the required correlations converge. The limiting correlations are averages of component correlations, called phases. One selected phase gives quantum fields on a Hilbert space, with positive energy, relativistic covariance and commutativity at spacelike separation. The same local product expansions hold in the average and in every phase. A logarithmic correction in a particular two-point function distinguishes it from the finite free-field expressions specified below. The result does not establish a mass gap.

Part I begins with definitions of the lattice model and its continuum objects, followed by the precise theorem. The strategy then separates the proof into five results: constructing the finite-volume limits, passing to infinite volume and its phases, reconstructing relativistic fields, expanding local products, and proving the logarithmic two-point law. Part II develops their proofs. It begins with local lattice observables, studies what happens when short-distance variables are integrated out, and then follows the resulting correlations into the continuum theory. This order keeps the finite probability model in view while explaining how its limiting correlations become quantum fields.

The statements are matched to the fixed formal-source revision in [1]. The present sections are a revision for review; the detailed arguments still missing from Part II are identified explicitly where they are used.

2 Definitions

We begin with the finite-dimensional lattice model and the observables whose averages will converge. We then specify the continuum correlations, their decomposition into phases and the relativistic fields reconstructed from one phase. The last definitions give the two short-distance tests in the theorem. Throughout, $\mathbb{N} = \{0, 1, \dots\}$; empty products equal 1, and correlation functions of order zero are included.

2.1 The lattice and its measure

Definition 2.1 (Gauge group)**.** A gauge group consists of an integer $m \geq 1$, a closed subgroup $G \subset U(m)$ and a real linear subspace $\mathfrak{g} \subset M_m(\mathbb{C})$ such that

$$X \in \mathfrak{g} \quad \Longleftrightarrow \quad \exp(sX) \in G \text{ for every } s \in \mathbb{R}\,. \tag{2.1}$$

Its Lie algebra $\mathfrak{g}$ is simple if $[X, Y] = XY - YX \neq 0$ for some $X, Y \in \mathfrak{g}$, and every real subspace $J \subset \mathfrak{g}$ with $[\mathfrak{g}, J] \subset J$ equals 0 or $\mathfrak{g}$.

The matrices in G will be the variables of the lattice model. The specified embedding in $U(m)$ fixes the trace used in its energy. We do not assume that G is connected.

Definition 2.2 (Periodic lattice)**.** For G as in Definition 2.1 and a positive integer ℓ, let $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$ and write e_μ, $1 \leq \mu \leq 4$, for its coordinate unit vectors. A positive link (y, μ) joins y to $y + e_\mu$. A link configuration is a map

$$U : \Lambda_\ell \times \{1, 2, 3, 4\} \longrightarrow G\,. \tag{2.2}$$

The value assigned to the reversed link is $U(y, \mu)^{-1}$.

A link matrix describes parallel transport along one edge. Multiplying four of these matrices around an elementary square compares transport along its two coordinate paths. This product, the plaquette matrix, is the local quantity from which the Wilson energy is formed.

Definition 2.3 (Plaquettes and energy)**.** For a configuration as in Definition 2.2, set

$$\begin{aligned} U_{y;\mu\nu} &= U(y, \mu)U(y + e_\mu, \nu)U(y + e_\nu, \mu)^{-1}U(y, \nu)^{-1}, \\ E_\ell(U) &= \sum_{y \in \Lambda_\ell} \sum_{\mu < \nu} \left(1 - \frac{1}{m} \operatorname{Re} \operatorname{tr} U_{y;\mu\nu}\right)\,. \end{aligned} \tag{2.3}$$

The energy counts each unordered pair of coordinate directions once. Each summand lies in $[0, 2]$, so $0 \leq E_\ell \leq 12\ell^4$. The probability measure weights configurations according to this energy.

Definition 2.4 (Wilson measure)**.** Let G be as in Definition 2.1 and let h be normalized Haar measure on G. For $\beta \in \mathbb{R}$, the Wilson measure on the configurations of Definition 2.2 is

$$\begin{aligned} Z_{\ell,\beta} &= \int e^{-\beta E_\ell(U)} \prod_{y,\mu} h(dU(y, \mu)), \\ \mu_{\ell,\beta}(dU) &= \frac{e^{-\beta E_\ell(U)}}{Z_{\ell,\beta}} \prod_{y,\mu} h(dU(y, \mu))\,. \end{aligned} \tag{2.4}$$

The bound on E_ℓ makes $Z_{\ell,\beta}$ finite and strictly positive for every real β. Increasing β gives greater relative weight to configurations with smaller energy. Changing coordinates in the internal matrix space independently at each site acts as follows.

Definition 2.5 (Gauge transformations)**.** For configurations as in Definition 2.2 and $\gamma : \Lambda_\ell \to G$, define

$$U^\gamma(y,\mu) = \gamma(y)U(y,\mu)\gamma(y+e_\mu)^{-1}\,. \tag{2.5}$$

An observable F on the configurations of Definition 2.2 is gauge invariant if $F(U^\gamma) = F(U)$ for every U and γ.

2.2 Local observables and the continuum limit

An observable is a function of the link matrices; its expectation is a measurement in the finite lattice model. To pass to a field at a point, we use observables that depend on links near one site. Their coefficients may change as the lattice is refined.

Definition 2.6 (Lattice observable labels and insertions)**.** Let $\mathcal{T}$ be a set of labels for lattice observables. An insertion of label $v \in \mathcal{T}$ at $y \in \Lambda_\ell$ is a complex-valued function $U \mapsto I(v,y;U)$ on link configurations (Definition 2.2). For a finite set $\Sigma \subset \mathbb{Z}^4$, the insertion is supported on the links inside $y + (\Sigma \bmod \ell)$ if it takes the same value on any two configurations that agree on all links whose two endpoints belong to this set.

At this stage a label only names an observable; it need not yet name a continuum field. The map relating the two kinds of labels will be part of the continuum data. We next choose the physical sizes of the lattices. The index N refines the spacing and p increases the volume.

Definition 2.7 (Wilson family)**.** For a gauge group (Definition 2.1), a Wilson family $\mathcal{W}$ consists of strictly increasing positive integer sequences $(D_N)_{N\in\mathbb{N}}$ and $(L_p)_{p\in\mathbb{N}}$, real numbers $\beta_{N,p}$, normalized Haar measure h on G, a label set $\mathcal{T}$, and insertions $I_{N,p}(v,y;U)$ as in Definition 2.6 on $\Lambda_{L_pD_N}$. We write

$$a_N = \frac{1}{D_N}, \qquad \ell_{N,p} = L_pD_N, \qquad \mu_{N,p} = \mu_{\ell_{N,p},\beta_{N,p}}\,. \tag{2.6}$$

The lattice has L_pD_N sites per side, separated by physical distance a_N; thus its physical side is L_p. Taking $N \to \infty$ at fixed p refines the lattice without enlarging the box. Taking $p \to \infty$ enlarges the box. The family also specifies the observables whose averages are to have limits.

Definition 2.8 (Regularity)**.** A Wilson family (Definition 2.7) is regular if:

Reg1. There exists $b_* > 0$ such that, for every $\varepsilon > 0$, there exists N_0 with

$$\left| \frac{\beta_{N,p}}{\log\left(\frac{1}{a_N}\right)} - b_* \right| < \varepsilon \qquad (N \ge N_0,\ p \in \mathbb{N})\,. \tag{2.7}$$

Reg2. Each $U \mapsto I_{N,p}(v,y;U)$ is continuous.

Reg3. Each insertion is gauge invariant in Definition 2.5.

Reg4. For every $v \in \mathcal{T}$, there exists a finite set $\Sigma_v \subset \mathbb{Z}^4$ such that every $I_{N,p}(v,y;\cdot)$ is supported on the links inside $y + (\Sigma_v \bmod \ell_{N,p})$ in Definition 2.6.

The same finite set in Reg4 works at every spacing, volume and site. Its diameter in physical units tends to zero with a_N. Condition Reg1 is a logarithmic growth law for the bare inverse coupling, uniform in the volume; its positive slope is left unspecified here.

Definition 2.9 (Physical positions and moments)**.** Let $r_\ell(j)$ be the representative of $j \in \mathbb{Z}/\ell\mathbb{Z}$ in $\left(-\frac{\ell}{2}, \frac{\ell}{2}\right]$. The physical position of $y \in \Lambda_{\ell_{N,p}}$ is

$$\hat{y} = a_N (r_{\ell_{N,p}}(y^\mu))_{\mu=1}^4 . \tag{2.8}$$

For a Wilson family (Definition 2.7) and $v = (v_1, \dots, v_n) \in \mathcal{T}^n$, define the moment and its smearing by a function f by

$$\begin{aligned} M_{N,p}(v; y) &= \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, \mu_{N,p}(dU), \\ C_{N,p}(v; f) &= a_N^{4n} \sum_{y \in \Lambda^n_{\ell_{N,p}}} f(\hat{y}_1, \dots, \hat{y}_n) M_{N,p}(v; y) . \end{aligned} \tag{2.9}$$

A correlation is an average of a product of observables. Smearing averages its spatial arguments against a test function, so the lattice sum can be compared with a continuum integral. We test only configurations of distinct points.

Definition 2.10 (Separated tests)**.** For $n \in \mathbb{N}$, put

$$\mathrm{Conf}_n = \{x \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ whenever } i \neq j\} . \tag{2.10}$$

A separated test is a smooth function $f : (\mathbb{R}^4)^n \to \mathbb{C}$ whose closed support is compact and contained in Conf_n.

Such a test vanishes near every coincidence of two points. It can depend jointly on all n points and need not be a product of one-point tests. To compare a torus with Euclidean space, its support must also stay inside the physical box.

Definition 2.11 (Tests supported inside a volume)**.** A separated test f (Definition 2.10) is supported inside volume p if

$$\operatorname{supp} f \subset \left(\left(-\frac{L_p}{2}, \frac{L_p}{2} \right)^4 \right)^n . \tag{2.11}$$

Definition 2.12 (Ordered limit)**.** For a Wilson family (Definition 2.7), $P : \mathbb{N} \to \mathbb{N}$ and kernels $K_n(v; x)$, convergence along P means the following. For every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $P(k)$ eventually in k, and there exist $R_p \in \mathbb{C}$ such that

$$\begin{aligned} \lim_{N \to \infty} C_{N,p}(v; f) &= R_p \qquad\qquad (f \text{ supported in volume } p), \\ \lim_{k \to \infty} R_{P(k)} &= \int_{(\mathbb{R}^4)^n} f(x) K_n(v; x)\, dx . \end{aligned} \tag{2.12}$$

Here the moments are those of Definition 2.9 and the support condition is Definition 2.11. The first limit uses the entire spacing sequence at every volume containing the test. The second limit enlarges the volume along P. At volumes not containing the support, R_p is unconstrained. A strictly increasing P contains each fixed compact test support eventually, since $L_{P(k)} \to \infty$.

Definition 2.13 (Periodic separation)**.** For a Wilson family (Definition 2.7) and volume p, the periodically separated configurations are

$$\{x \in (\mathbb{R}^4)^n : x_i - x_j \notin L_p\mathbb{Z}^4 \text{ whenever } i \neq j\}\,. \tag{2.13}$$

Distinct points can represent the same point of a torus if their difference is a period. This definition excludes those coincidences as well, and is used for fixed-volume correlation kernels.

2.3 The continuum objects

Definition 2.14 (Spacetime and Schwartz tests)**.** Write $x = (\vec{x}, t(x)) \in \mathbb{R}^3 \times \mathbb{R}$ for spacetime, with $t(x) = x^4$, Euclidean norm $|x|$ and

$$\mathsf{m}(x) = t(x)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{x : |\vec{x}| \leq t(x)\}\,. \tag{2.14}$$

A vector is spacelike if $\mathsf{m}(x) < 0$. Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^4; \mathbb{C})$, with its Schwartz topology.

Hilbert inner products are antilinear in the first variable. The next definition lists the Hilbert space and fields before imposing their physical properties. A field is smeared against a Schwartz function because its value at a single point need not be an operator.

Definition 2.15 (Relativistic data)**.** A relativistic datum consists of a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$, a vector $\Omega \in \mathcal{D}$ and fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ linear over $\mathbb{C}$ in $\ell \in \mathcal{L}$ and $f \in \mathcal{S}$ (Definition 2.14). It includes an antilinear map $\ell \mapsto \ell^*$, a strongly continuous unitary translation representation $T(b)$ and a projection-valued Borel measure E on $\mathbb{R}^4$. Its diagonal finite measures satisfy $\mu_\psi(A) = \langle \psi, E(A)\psi \rangle$ on Borel sets.

There is also a strongly continuous unitary representation U of a topological group $\mathcal{P}$, identified by a homeomorphism κ with the proper orthochronous Poincaré group. Writing $\kappa(g) = (\Lambda_g, b_g)$, this identification respects identity and the multiplication

$$(\Lambda, b)(\Lambda', b') = (\Lambda\Lambda', \Lambda b' + b)\,. \tag{2.15}$$

Here Λ is invertible, has determinant 1, preserves m and maps $\overline{V}_+$ onto itself. Pure translations $\tau(b) = \kappa^{-1}(1, b)$ satisfy $U(\tau(b)) = T(b)$. For each $g \in \mathcal{P}$ there is a complex linear bijection $\rho(g)$ of $\mathcal{L}$. The action on tests is $(g \cdot f)(x) = f(\Lambda_g^{-1}(x - b_g))$.

A projection-valued Borel measure here is normalized by $E(\mathbb{R}^4) = 1$, has $E(\varnothing) = 0$, satisfies $E(A)E(B) = E(A \cap B)$ and $E(A)^* = E(A)$ on Borel sets, and is countably additive in the strong operator topology. One can assign arbitrary operators to non-Borel sets, which none of the assertions uses. The label map * is not required to be an involution. No continuity or finite-dimensionality condition on ρ is included in the data.

Definition 2.16 (Wightman conditions)**.** The relativistic datum of Definition 2.15 satisfies the following conditions, with spacetime and test functions as in Definition 2.14 and $p \cdot b$ the Euclidean pairing.

W0. For every $\psi \in \mathcal{H}$ and $b \in \mathbb{R}^4$,

$$\langle \psi, T(b)\psi \rangle = \int e^{ip\cdot b}\, \mu_\psi(dp), \qquad E(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,. \tag{2.16}$$

Every vector fixed by all $T(b)$ belongs to $\mathbb{C}\Omega$.

W1. The vacuum has norm one, $\mathcal{D}$ is dense, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_j, f_j)\Omega : j \in \mathbb{N},\ \ell_i \in \mathcal{L},\ f_i \in \mathcal{S}\}\,. \tag{2.17}$$

For every $\ell \in \mathcal{L}$ and $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ into $\mathcal{H}$ in norm. For every ℓ, f and $\psi, \chi \in \mathcal{D}$,

$$\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle\,. \tag{2.18}$$

W2. The maps ρ form a representation: $\rho(1) = 1$ and $\rho(gg') = \rho(g)\rho(g')$. For every $g \in \mathcal{P}$, $U(g)\mathcal{D} \subset \mathcal{D}$ and $U(g)\Omega = \Omega$. For every g, ℓ, f and $\psi \in \mathcal{D}$,

$$U(g)\Phi(\ell, f)\psi = \Phi(\rho(g)\ell, g \cdot f)U(g)\psi\,. \tag{2.19}$$

W3. If $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$ and $y \in \operatorname{supp} f'$, then, for every $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$,

$$\Phi(\ell, f)\Phi(\ell', f')\psi = \Phi(\ell', f')\Phi(\ell, f)\psi\,. \tag{2.20}$$

The word span is the domain itself, and includes the vacuum as the empty word. The symmetry group is the Poincaré group itself rather than its double cover. The sign convention in (2.16) is converted to the usual Minkowski pairing by reversing spatial momentum; the forward cone is unchanged.

We collect the continuum objects in a hierarchy: a family of correlations, their phase decomposition, the relativistic fields, and the coefficients for local product expansions. The state mix denotes the average supplied by the lattice limit; a state $\xi \in X$ denotes one phase.

Definition 2.17 (Hierarchy)**.** A hierarchy over the Wilson family of Definition 2.7 consists of the following data.

H1. A relativistic datum of Definition 2.15.

H2. For every $d \in \mathbb{N}$, a finite-dimensional complex normed space F_d and a complex linear map $\iota_d : F_d \to \mathcal{L}$. Set $\mathbb{F} = \bigsqcup_{d\in\mathbb{N}} F_d$, $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$.

H3. A map $\pi : \mathcal{T} \to \mathbb{F}$, assigning a continuum field to each lattice observable label (Definition 2.6).

H4. A compact metric space X, called the phase space, with a Borel measure ν and a selected point $\xi_* \in X$.

H5. For every $n \in \mathbb{N}$ and $s \in X \sqcup \{\mathrm{mix}\}$, a correlation function $S^s(a; x) \in \mathbb{C}$ on $\mathbb{F}^n \times (\mathbb{R}^4)^n$.

H6. For every $n \in \mathbb{N}$, a continuation $\mathsf{F}_n(a; z, \vec{x}) \in \mathbb{C}$ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$.

H7. For every $j \in \mathbb{N}$, a finite index set Ch_j and fields $O_{j,\kappa} \in F_j$. For every finite index set I, $a \in \mathbb{F}^I$ and $\kappa \in \mathrm{Ch}_j$, there is a coefficient function $c_{j,\kappa}(a; x, o)$ on $(\mathbb{R}^4)^I \times \mathbb{R}^4$.

H8. An element $\Theta \in F_4$.

The degree is an index used in the product estimates below. The definition imposes no inclusions between different F_d, no prescribed curvature interpretation and no compatibility of the ι_d across degrees. The functions S^s are defined even on collision diagonals, but the conclusions constrain them only at separated points.

2.4 Reconstruction and phases

The mixture will be the limit of lattice moments. The point ξ_* instead selects the Euclidean correlations that continue to the relativistic fields. These two roles are related by a phase average, not by identifying the two states.

Definition 2.18 (Reconstruction properties)**.** A hierarchy of Definition 2.17 has the reconstruction properties if the following hold. Here $s \in X \sqcup \{\mathrm{mix}\}$, $n \in \mathbb{N}$ and $a \in \mathbb{F}^n$.

R1. Its relativistic datum satisfies W0–W3.

R2. For every nonzero $\ell \in \mathcal{L}$, there exists $f \in \mathcal{S}$ with $\Phi(\ell, f) \neq 0$.

R3. Each ι_d is injective.

R4. Some $a \in \mathbb{F}$ and $f \in \mathcal{S}$ satisfy $\Phi(\iota(a), f) \neq 0$.

R5. The map $a \mapsto \iota(a)$ from $\mathbb{F}$ onto $\mathcal{L}$ is surjective.

R6. The map $\pi : \mathcal{T} \to \mathbb{F}$ is surjective.

R7. The measure ν has total mass one.

R8. Every nonempty open subset of X has positive ν-measure.

R9. For every state s, order n, list $a \in \mathbb{F}^n$, $x \in \mathrm{Conf}_n$ and permutation σ,

$$S^s((a_{\sigma(i)})_i; (x_{\sigma(i)})_i) = S^s(a; x)\,. \tag{2.21}$$

R10. For every s, n, a, the function $S^s(a; \cdot)$ is smooth on Conf_n.

R11. For every n, a and $x \in \mathrm{Conf}_n$, the function $\xi \mapsto S^\xi(a; x)$ is continuous.

R12. For every n, a and $x \in \mathrm{Conf}_n$,

$$\int_X S^\xi(a; x)\, \nu(d\xi) = S^{\mathrm{mix}}(a; x)\,. \tag{2.22}$$

R13. For every n, a and fixed real spatial configuration $\vec{x}$, the function $z \mapsto \mathsf{F}_n(a; z, \vec{x})$ is holomorphic on

$$\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}\,. \tag{2.23}$$

R14. For every n, a and real configuration x with $t(x_1) < \cdots < t(x_n)$,

$$\mathsf{F}_n(a; (it(x_i))_i, (\vec{x}_i)_i) = S^{\xi_*}(a; x)\,. \tag{2.24}$$

R15. For every n, a, real heights $h_1 < \cdots < h_n$ and tests $f_1, \ldots, f_n \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int_{(\mathbb{R}^4)^n} \mathsf{F}_n(a; (t(x_i) + irh_i)_i, (\vec{x}_i)_i) \prod_{i=1}^n f_i(x_i)\, dx = \langle \Omega, \Phi(\iota(a_1), f_1) \cdots \Phi(\iota(a_n), f_n)\Omega \rangle\,. \tag{2.25}$$

In (2.25) we retain the formal source's convention that the integral of a nonintegrable function is zero. No separate positive-height integrability assertion is part of R15. The integrals in the lattice limit and (2.22) are ordinary absolutely convergent integrals, by compact support and smoothness in the first case and by phase continuity and compactness in the second.

2.5 Products and the trace test

The product expansion replaces a cluster by finitely many single fields at its center. Fields whose positions remain away from the cluster are called spectators. One choice of coefficients must work with every list of such fields and in every state. The threshold on the cluster radius, however, may depend on the fixed spectator fields.

Definition 2.19 (Cluster geometry)**.** For a hierarchy (Definition 2.17) and finite index sets I, J, write $S^s(a, x; b, y)$ for the correlation on the concatenated lists, enumerated in one fixed order on $I \sqcup J$. For $r > 0$, $M \geq 0$ and $\eta > 0$, call (x, o) an (r, M, η)-cluster if

$$|x_i - o| \leq Mr \quad (i \in I), \qquad |x_i - x_{i'}| \geq \eta r \quad (i \neq i')\,. \tag{2.26}$$

For $c > 0$, call y c-clear of o if

$$|o - y_j| \geq c \quad (j \in J), \qquad |y_j - y_{j'}| \geq c \quad (j \neq j')\,. \tag{2.27}$$

Set $w(a) = \sum_{i \in I} \deg a_i$.

Definition 2.20 (Operator product expansion)**.** For a hierarchy (Definition 2.17), the OPE property consists of the following estimates on clusters and distant points as in Definition 2.19.

O1. For every finite I, $a \in \mathbb{F}^I$, $M \geq 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\kappa \in \mathrm{Ch}_j$, there exists $\delta > 0$ such that, for every $0 < r < \delta$ and every (r, M, η)-cluster,

$$|c_{j,\kappa}(a; x, o)| \leq r^{\,j - w(a) - \varepsilon}\,. \tag{2.28}$$

O2. For every finite I, J, $a \in \mathbb{F}^I$, $b \in \mathbb{F}^J$, $M \geq 0$, $\eta, c, \varepsilon > 0$ and $D \in \mathbb{N}$, there exists $\delta > 0$ such that, for every $0 < r < \delta$, every (r, M, η)-cluster, every c-clear spectator configuration and every $s \in X \sqcup \{\mathrm{mix}\}$,

$$\left| S^s(a, x; b, y) - \sum_{j=0}^{D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(a; x, o)\, S^s((j, O_{j,\kappa}), o; b, y) \right| \leq r^{\,D+1-w(a)-\varepsilon}\,. \tag{2.29}$$

The coefficient functions are those already chosen in H7. They do not depend on spectators, state or truncation degree. Both estimates have constant 1, with the dependence absorbed into δ. For small enough r, $Mr < c$ ensures that all points in the tested correlations are separated. These are finite-truncation estimates; they assert neither an infinite convergent expansion nor uniqueness of its coefficients.

Definition 2.21 (The trace two-point function)**.** For the field $\Theta \in F_4$ of Definition 2.17, set

$$G_\Theta(r) = S^{\xi_*}((4, \Theta), (4, \Theta); -\frac{r}{2} e_4, \frac{r}{2} e_4)\,. \tag{2.30}$$

Free-field two-point functions are built from the energy $E_{\mathsf{m}}(p)$ of a particle with mass m and spatial momentum p. The following class allows finitely many masses and derivatives, and finite sums of products of the resulting kernels. It specifies exactly the comparisons excluded by the theorem.

Definition 2.22 (Finite Wick profiles)**.** A mode consists of $\mathsf{m} \geq 0$, integers $j, q \in \mathbb{N}$ and a continuous real function h on $\mathbb{R}^3$ satisfying $h(\lambda p) = \lambda^j h(p)$ for every $\lambda > 0$. Set

$$E_{\mathsf{m}}(p) = \sqrt{\mathsf{m}^2 + |p|^2}, \qquad K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} \frac{E_{\mathsf{m}}(p)^q h(p)}{2E_{\mathsf{m}}(p)} e^{-rE_{\mathsf{m}}(p)}\, dp\,. \tag{2.31}$$

The integrand can be assigned value zero at a zero denominator, a set of measure zero. A finite Wick profile is

$$W(r) = \sum_{\alpha=1}^{A} c_\alpha \prod_{\beta=1}^{n_\alpha} K_{\mathsf{m}_{\alpha\beta},q_{\alpha\beta},h_{\alpha\beta}}(r), \qquad c_\alpha \in \mathbb{C}, \quad A, n_\alpha \in \mathbb{N}\,. \tag{2.32}$$

Empty products are allowed.

Definition 2.23 (Trace test)**.** The hierarchy of Definition 2.17 passes the trace test if the following hold, with G_Θ as in Definition 2.21 and Wick profiles as in Definition 2.22.

T1. For every $f \in \mathcal{S}$, $\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0$.

T2. The number $A_G = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}$ is positive.

T3. The following limit holds in $\mathbb{C}$ over all positive radii:

$$\lim_{r\downarrow 0} r^8 \left(\log \frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}\,. \tag{2.33}$$

T4. For every finite Wick profile W and every $\delta > 0$, there exists $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$.

The public trace test concerns the chosen degree-four field and the time axis. Its definition alone supplies no stress tensor. The construction identifies this field with a centered stress trace, as used in the proof of Proposition 5.5 below.

3 The main theorem

The theorem ties the continuum correlations to the original Wilson measures. The lattice family is chosen once; the later choice of a volume subsequence determines one continuum hierarchy with all four conclusions below.

Theorem A. *For every gauge group $(m, G, \mathfrak{g})$ with $\mathfrak{g}$ simple (Definition 2.1), there exists a Wilson family $\mathcal{W}$ (Definition 2.7) satisfying the regularity conditions Reg1–Reg4 of Definition 2.8 such that, for every strictly increasing $p : \mathbb{N} \to \mathbb{N}$, there exist a strictly increasing $q : \mathbb{N} \to \mathbb{N}$ and a hierarchy over $\mathcal{W}$ (Definition 2.17) with the following properties.*

P1. *For every $n \in \mathbb{N}$, $v \in \mathcal{T}^n$ and separated test f (Definition 2.10), the test is supported inside volume $p(q(k))$ eventually in k (Definition 2.11), and there exist $R_j \in \mathbb{C}$ such that*

$$\begin{aligned} \lim_{N\to\infty} C_{N,j}(v; f) &= R_j && (f \text{ supported in volume } j), \\ \lim_{k\to\infty} R_{p(q(k))} &= \int f(x) S^{\mathrm{mix}}(\pi(v_1), \dots, \pi(v_n); x)\, dx\,. \end{aligned} \tag{3.1}$$

Here $C_{N,j}$ is the Wilson moment of Definition 2.9; these are the ordered limits of Definition 2.12.

P2. *The selected relativistic datum (Definition 2.15) satisfies the Wightman conditions W0–W3 of Definition 2.16. Its field labels are faithful, each ι_d is injective, both π and $a \mapsto \iota(a)$ are surjective, and some $\Phi(\iota(a), f)$ is nonzero, as in R2–R6. The measure ν is a probability measure of full support; all the S^s are symmetric and smooth at separated points, their phase values are continuous, and*

$$S^{\mathrm{mix}}(a;x) = \int_X S^{\xi}(a;x)\,\nu(d\xi) \qquad (x \in \mathrm{Conf}_n)\,.$$

The continuation F_n is holomorphic in ordered complex times, equals S^{ξ_*} at ordered imaginary times, and has the selected vacuum expectations as its product-test boundary values. These assertions have exactly the domains and quantifiers in R1–R15 of Definition 2.18.

P3. *The single coefficient system in H7 satisfies the coefficient estimate (2.28) and the remainder estimate (2.29) of Definition 2.20, for all finite truncations, all allowed clusters and distant points (Definition 2.19), and every state $s \in X \sqcup \{\mathrm{mix}\}$, with the thresholds in O1–O2.*

P4. *The field $\Theta \in F_4$ has zero vacuum expectation, and its selected-phase two-point function (Definition 2.21) satisfies*

$$\lim_{r\downarrow 0} r^8 \left(\log \frac{1}{r}\right)^2 G_\Theta(r) = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4} > 0\,.$$

For every finite Wick profile W (Definition 2.22) and every $\delta > 0$, there exists $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$. These are the trace-test conditions T1–T4 of Definition 2.23.

The lattice data are chosen before p. The refinement, field spaces, phases, selected relativistic theory, product coefficients and trace field may depend on p. The same refinement works for all orders of correlation, observable labels and separated tests. The lattice limit in P1 is the mixture; the relativistic theory in P2 is attached to ξ_*.

The theorem does not assert uniqueness of the infinite-volume limit, universality with respect to lattice data, or convergence along simultaneous cutoff and volume limits. It does not assert a mass gap, clustering of the mixture, agreement of phases, or that an individual phase is a lattice-state limit. It gives no contact extension across collisions, and excludes only the finite profiles of Definition 2.22, not all composites of Gaussian fields or continuous-mass generalized free fields. No Euclidean covariance or reflection-positivity assertion is included among P1–P4; those properties are established internally in the construction.

4 Strategy

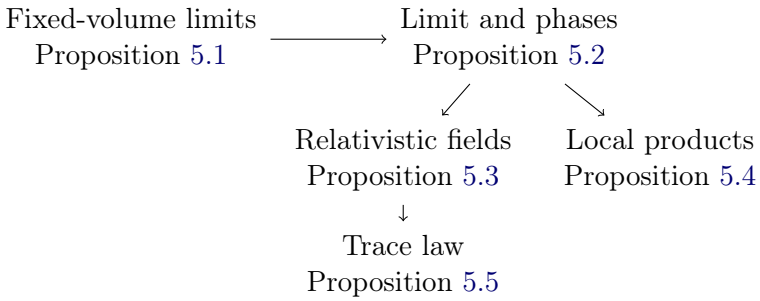
The proof has two limiting operations and then three ways of reading the same limiting correlations. First we remove the lattice spacing while keeping the physical box fixed. Then we enlarge the box, allowing a subsequence of volumes. The later arguments recover relativistic fields, compare products at nearby points, and distinguish a two-point function from the specified free-field profiles. None of them may replace the correlations obtained in the first two steps.

From small spacings to a fixed box. Refining the lattice adds new fluctuations at shorter distances. We integrate these fluctuations a few scales at a time, adjusting the bare coupling so that the effective coupling at a fixed reference scale is always the same. The observables must

be adjusted along with the probability law; otherwise their correlations need not remain finite. Proposition 5.1 produces one family of observables with these adjustments and proves their limits at each fixed volume. Its estimates are local, so they can later control regions inside arbitrarily large boxes.

From boxes to phases. Local bounds independent of box size allow a subsequence on which the correlations converge. This is the compactness step in Proposition 5.2. The result can still be an average of different states. Reflection positivity turns its positive-time correlations into inner products, and averages at widely separated times distinguish its components. After continuous rotations have been established, these averages commute and can be represented as functions on a compact space of phases. Their expectation is integration against a probability measure, giving the mixture formula (2.22). The phase construction is therefore an additional argument after compactness.

Three conclusions from one hierarchy. Proposition 5.3 selects a phase with a unique vacuum and continues its time variables to obtain relativistic fields. Proposition 5.4 replaces a product at nearby points by a finite sum of fields at one point, with a controlled error inside any correlation with distant fields. Its coefficients are chosen once, so the expansion also holds in every phase. Proposition 5.5 studies the centered trace of the stress tensor: a logarithmic correction to its two-point singularity rules out the finite Wick profiles. The dependence of these five results is summarized by



Here the last three results all use estimates for the same lattice family and the same continuum hierarchy supplied by the first two. The arrows show the order of construction, not a claim that compactness alone gives their analytic inputs.

For each gauge group, we make one joint choice of the family in Proposition 5.1. For each prescribed increasing volume sequence, we then make one joint choice of a refinement and hierarchy for Propositions 5.2–5.5. Their conclusions are not asserted for an arbitrary regular family or an arbitrary smooth table of correlations. With these common choices, the deduction of Theorem A is just the assembly of the five conclusions; it appears at the end of the next section.

5 The principal propositions

The five propositions below use the joint choices described in Section 4. Their complete proofs belong to Part II; the explanations here give the ideas needed to follow the deduction of Theorem A.

Removing the lattice spacing. At a fixed physical volume, increasing the number of sites must not change what the model looks like at a fixed larger distance. The coupling and the normalization

of each observable are chosen to achieve this compatibility. The first proposition gives both one such choice and convergence along its entire spacing sequence.

Proposition 5.1 (Ultraviolet construction)**.** *For every gauge group with simple* $\mathfrak{g}$ *(Definition 2.1), there exists a regular Wilson family* $\mathcal{W}$ *(Definitions 2.7 and 2.8) such that, for every* $p, n \in \mathbb{N}$ *and* $v \in \mathcal{T}^n$*, there exists a smooth kernel* $K_{p,v}$ *on the periodically separated set of Definition 2.13 with*

$$\lim_{N\to\infty} C_{N,p}(v; f) = \int f(x) K_{p,v}(x)\, dx \tag{5.1}$$

for every separated test f *supported inside volume* p *(Definitions 2.9, 2.10 and 2.11).*

Idea of the proof. Replace the fine link variables by averaged variables at a larger spacing, keeping the integral of the probability density exactly the same. Where the fields are small, their fluctuations can be compared with a Gaussian integral. Where they are large, the energy suppresses their contribution; these configurations are estimated rather than discarded. Repeating this integration changes the coefficient multiplying the squared curvature. We account for that change in the coupling, instead of treating it as a small error. The resulting coupling grows logarithmically as the original spacing shrinks.

We carry the local observables through the same integrations. Their normalization is fixed at a reference scale, so different cutoffs are compared there rather than at their different bare couplings. To prove convergence, stop the integration at an intermediate scale. At this fixed scale the effective correlations converge with the cutoff, while the remaining error is uniformly small when the intermediate scale is sufficiently fine. Choosing the intermediate scale first and the cutoffs second proves the Cauchy property.

Unresolved. The later ultraviolet sections must prove the estimates for integrating small and large fields, identify the change of the actual effective coupling, and establish the fixed-scale Cauchy estimate with a uniformly small remaining error. The fixed-volume conclusion has been checked in the formal source; its analytic proof is still missing from this manuscript.[1]

Enlarging the volume and separating phases. The estimates for a fixed collection of observables must remain bounded when the surrounding box grows. They give compactness, but the limit can retain several possible vacuum states. The next proposition includes the separate argument that resolves this average into phases.

Proposition 5.2 (Continuum limit and phases)**.** *For the jointly chosen family of Proposition 5.1 and every strictly increasing* $p : \mathbb{N} \to \mathbb{N}$*, there exist a strictly increasing* $q : \mathbb{N} \to \mathbb{N}$ *and hierarchy data (Definition 2.17) satisfying the limit P1 and the phase and regularity conditions R7–R12 of Definition 2.18.*

Idea of the proof. Reflection positivity bounds an average of several separated observables by averages of reflected copies of each individual observable. The ultraviolet estimates bound these individual averages independently of box size. A positive transfer operator controls changes in the correlations when their points move. Boundedness and equicontinuity then allow a subsequence to be selected for all finite label bases and all compact sets at once. Lattice summation by parts relates derivatives of the limits to insertions of higher degree and gives smoothness. The fixed-volume

[1] Formal correspondence at revision `3456e5e`: `Construction.lean`, lines 15–40; `HierarchyBridge.lean`, lines 16–26; `YM/FullVolume/Profiles.lean`, lines 25–47..

limits of Proposition 5.1 identify every inner limit, so only the volume subsequence remains to be chosen.

The average state is split into phases by examining observables separated by a very long time. In the Hilbert space obtained from reflection positivity, long-time evolution removes the positive-energy part and retains the zero-energy vectors. Observables placed between these long evolutions act on that zero-energy space. Once rotations are established, exchanging a time direction with a spatial direction shows that these limiting operators commute. The spectral representation of their commutative algebra gives the compact phase space; the vacuum expectation gives its probability measure. This produces the mixture identity (2.22).

Unresolved. The infinite-volume argument still needs the same-family bounds for local observables and their derivative representatives from the ultraviolet sections. The later symmetry and phase sections must also prove that the contribution from a far-away stress integral tends to zero and that the long-time operators commute; neither follows from compactness alone.[2]

Recovering relativistic fields. Euclidean time evolution damps positive energies. Allowing time to become complex relates this damping to unitary time evolution, and hence to relativistic vacuum expectations. The estimates must hold as the imaginary time separations tend to zero, because the resulting fields are distributions.

Proposition 5.3 (Relativistic reconstruction)**.** *For the jointly chosen hierarchy of Proposition 5.2, the selected theory satisfies R1–R6 and the continuation satisfies R13–R15 of Definition 2.18, with the same fields and phase as in Definition 2.17.*

Idea of the proof. First place Euclidean observables in thin, separated time intervals. Their bounds show that the continued correlations have distributional limits when the imaginary time separations disappear. Positivity turns these limits into inner products of field words applied to a vacuum. Completing this space gives the Hilbert space, and adding a letter to a word defines a field operator.

Rotation identities continue to Lorentz transformations. Together with positivity of Euclidean energy, this confines energy and momentum to the forward cone. To prove locality, compare the two orders of neighboring fields at spacelike separated points: their analytic continuations agree on the Euclidean configurations and therefore give the same boundary values there. A label is discarded only if every correlation containing it vanishes. This quotient makes the remaining field labels faithful, while the unit supplies a nonzero field.

Unresolved. The reconstruction section must prove the bounds as imaginary time separations vanish and justify the equality of the two analytic orderings at spacelike separation. The exact selected-phase and product-test conclusions have been checked in the formal source; these analytic arguments remain unwritten.[3]

Replacing a cluster by a single field. A distant observation cannot resolve every detail of several nearby insertions. The operator product expansion expresses their effect through a finite list of fields at the cluster center. The coefficients must be independent of which distant observations are subsequently made.

[2]Formal correspondence at revision `3456e5e`: `YM/Extraction/Limits.lean`, lines 26–43; `HierarchyBridge.lean`, lines 44–57 and 82–119; the phase representation is in `YM/Phase/Mixture.lean`..

[3]Formal correspondence at revision `3456e5e`: `PublicTheory.lean`, lines 93–319; `HierarchyBridge.lean`, lines 82–133; boundary and locality proofs are organized in `YM/Work/C01.lean` through `C29.lean`..

Proposition 5.4 (Common operator product expansion)**.** *The fields and coefficients in H7 of the jointly chosen hierarchy satisfy O1–O2 of Definition 2.20, for the clusters of Definition 2.19.*

Idea of the proof. Average the clustered observables from their separation scale toward a larger scale, keeping track of which single local fields they produce. Retain the terms through degree D and estimate everything else. The first omitted degree gives the power $D+1$ in the remainder. The arbitrarily small loss in that power accommodates logarithmic factors. A fixed collection of correlations with distant test fields determines the coefficients once, before any later spectator fields or phase are chosen. The error is bounded as an operator in the same commutative algebra that defines the phases, so its bound holds in every phase and in the average state.

Unresolved. The product section must prove the local-product remainder estimate and show how the fixed coefficient choice gives an operator bound before evaluation in a phase. The quantifiers and both powers of the radius in O1–O2 have been checked; their analytic proof is still deferred.[4]

Detecting the logarithmic correction. The construction supplies a stress tensor and subtracts the selected vacuum expectation from its trace. Comparison at short distances identifies the leading two-point singularity of this centered field. The logarithmic correction to that singularity is incompatible with the finite free-field expressions in Definition 2.22.

Proposition 5.5 (Trace law and finite-profile exclusion)**.** *The field $\Theta \in F_4$ of the jointly chosen hierarchy satisfies T1–T4 of Definition 2.23 in the theory of Proposition 5.3, with the two-point function of Definition 2.21.*

Idea of the proof. At small distances the fields can be compared, after adjusting their normalizations, with free Maxwell fields. The trace carries an additional inverse running coupling at each insertion. Since that coupling grows logarithmically, its two-point function acquires the inverse square logarithm in (2.33). The coefficient of the logarithmic growth enters both the running coupling and the trace normalization, and cancels from their ratio. The final amplitude by itself therefore does not check that coefficient.

The comparison initially works at specially chosen scales. A positive spectral representation makes the two-point function decrease with the time separation, so its values at separations just above and below a requested one bound its value there. Every sequence of radii approaching zero has a subsequence where the comparison applies; thus a sequence contradicting the full limit cannot exist. Finite Wick profiles instead have expansions in integer powers of r and nonnegative powers of $\log r$. Such expansions cannot have the positive inverse-logarithmic behavior in (2.33).

Unresolved. The trace section must prove the stress normalization, comparison along subsequences of arbitrary radii, the spectral comparison between nearby separations, and the finite mass-shell expansion. The amplitude and the exact finite comparison class have been checked; these proofs remain to be supplied.[5]

Deduction of Theorem A from the principal propositions. Fix the gauge group and choose the family of Proposition 5.1, including its insertions and positive regularity slope. For a prescribed strictly increasing p, take the common q and hierarchy of Proposition 5.2. This gives P1 and R7–R12 on that refinement. Proposition 5.3 supplies the remaining reconstruction properties, hence P2. Propositions 5.4 and 5.5 give P3 and P4 for those same objects. □

[4] Formal correspondence at revision `3456e5e`: `ChallengeSpec.lean`, lines 58–80; `HierarchyBridge.lean`, lines 68–80; `YM/OPE/Orbits/Kernel.lean`, lines 62–77..

[5] Formal correspondence at revision `3456e5e`: `TraceBridge.lean`, lines 14–44; `YM/TraceExtension/Verse/Cape.lean`, lines 53–88; `YM/TraceExtension/Wick/Log.lean`, lines 96–108..

Part II
The construction

6 Lattice fields

A lattice observable measures the field through a finite collection of nearby links. To compare the same observable at finer spacings, we must specify both the classical quantity it represents and how its coefficients change with scale. We first prove the exact symmetries of the Wilson measure, then describe these two normalizations. The lattice, plaquettes and measure are those of Definitions 2.2, 2.3 and 2.4; their definitions will not be repeated here. Throughout, the gauge group is as in Definition 2.1, without a simplicity or connectedness assumption.

6.1 The Wilson measure

The action depends on the transport around each elementary square, or plaquette, as explained in Definition 2.3. Conjugating that transport or reversing its direction leaves the action unchanged. This observation gives exact symmetries of the probability measure, before any limit is taken.

Write $U(w)$ for the ordered transport along a walk w from x to z, using the link convention of Definition 2.2. The gauge action of Definition 2.5 gives

$$U^{\gamma}(w) = \gamma(x)U(w)\gamma(z)^{-1}\,. \tag{6.1}$$

The factors at intermediate vertices cancel, for forward and backward steps alike. For the estimates below, abbreviate the single-plaquette action and define its associated distance from the identity by

$$s(u) = 1 - \frac{1}{m}\operatorname{Re}\operatorname{tr} u, \qquad \chi(u) = \frac{\|u-1\|_{\mathrm{F}}}{\sqrt{m}} \qquad (u \in G), \tag{6.2}$$

where $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm. Unitarity gives

$$\chi(u)^2 = 2s(u), \qquad 0 \le s(u) \le 2, \qquad \chi(uv) \le \chi(u) + \chi(v)\,. \tag{6.3}$$

Indeed, $\|u-1\|_{\mathrm{F}}^2 = 2m - 2\operatorname{Re}\operatorname{tr} u$ and $\|u-1\|_{\mathrm{F}} \le 2\sqrt{m}$. For the last inequality, write $uv - 1 = (u-1) + u(v-1)$ and use invariance of the Frobenius norm under unitary multiplication. Both s and χ are continuous and invariant under conjugation and inversion.

The coordinate symmetries we use are the permutations and reversals of the four lattice axes. They act on link configurations by transporting each link along its image.

Definition 6.1 (Coordinate symmetries)**.** Let H_4 be the group of signed permutation matrices. For $A(y) = hy + z$, with $h \in H_4$ and $z \in \Lambda_\ell$, the pullback A^*U reads the transport along the image of each positive link. A reversed image uses the inverse variable, with its initial vertex shifted accordingly. The lattice and link conventions are those of Definition 2.2.

Write $\mathcal{U}_\ell$ for the configuration space of Definition 2.2, and $B_b(\mathcal{U}_\ell;\mathbb{C})$ for its bounded Borel complex functions. The following proposition justifies ordinary expectations of local polynomials and the changes of variables used throughout the construction.

Proposition 6.2 (The lattice law)**.** *For $\ell \ge 1$ and $\beta \in \mathbb{R}$, the measure of Definition 2.4 satisfies*

$$\mu_{\ell,\beta}(\mathcal{U}_\ell) = 1, \qquad C(\mathcal{U}_\ell;\mathbb{C}) \cup B_b(\mathcal{U}_\ell;\mathbb{C}) \subseteq L^1(\mu_{\ell,\beta})\,. \tag{6.4}$$

For every gauge transformation γ of Definition 2.5, coordinate symmetry A of Definition 6.1, and $F \in L^1(\mu_{\ell,\beta})$,

$$\int F(U^\gamma)\, d\mu_{\ell,\beta}(U) = \int F(U)\, d\mu_{\ell,\beta}(U) = \int F(A^*U)\, d\mu_{\ell,\beta}(U)\,. \tag{6.5}$$

Proof. The closed subgroup $G \subset U(m)$ is compact, so $\mathcal{U}_\ell$ is compact. The energy in (2.3) is continuous and satisfies $0 \le E_\ell \le 12\ell^4$, by (6.3) and the six increasing axis pairs. For every real β, the unnormalized density in (2.4) is bounded above and bounded below by a positive constant. Its integral is therefore finite and positive, and division by this integral proves normalization. Continuity on the compact configuration space implies boundedness, which proves both integrability assertions.

Probability Haar measure on a compact group is invariant under right translations and inversion as well as left translations. For example, its image under a fixed right translation is again left invariant with total mass one, so Haar uniqueness gives equality. Inversion exchanges left and right invariance and preserves the normalization. Each gauge transformation acts on a link by fixed left and right multiplications. It preserves product Haar measure, while (6.1) conjugates every plaquette transport at its initial vertex. Hence it preserves the energy and the density in (2.4).

An affine signed permutation permutes the positive-link indices and inverts selected variables, so it also preserves product Haar measure. The image of an elementary square is another elementary square; changing its initial corner conjugates its transport, and reversing its circulation inverts it. To check the indices, write $he_\mu = \sigma_\mu e_{\pi(\mu)}$, with $\sigma_\mu \in \{-1, 1\}$. The image square has its positive-axis base at

$$hy + z - \mathbf{1}_{\{\sigma_\mu = -1\}} e_{\pi(\mu)} - \mathbf{1}_{\{\sigma_\nu = -1\}} e_{\pi(\nu)}$$

and its axis pair is the increasing rearrangement of $\{\pi(\mu), \pi(\nu)\}$. The axis pairs are permuted, and for each pair the site map is an affine bijection. Thus the face indices are permuted even when the periodic side is one or two. Conjugation and inversion invariance of s give invariance of E_ℓ. Changing variables in (2.4) proves (6.5). □

6.2 Classical labels

We now choose concrete classical quantities for the lattice observable labels of Definition 2.6. A connection near the origin is described in radial gauge: its component in the radial direction vanishes. Its Taylor coefficients then satisfy a linear constraint, degree by degree. The constant coefficient vanishes: a constant one-form with zero contraction against every radial vector must be zero. A polynomial in finitely many such coefficients describes a classical local observable; invariance under simultaneous conjugation removes the choice of gauge at the origin.

Definition 6.3 (Classical labels)**.** For $\mathfrak{g}$ of Definition 2.1 and $j \ge 0$, let $\mathcal{A}_j$ be the real space of homogeneous polynomial one-forms of degree $j+1$, with values in $\mathfrak{g}$, satisfying

$$\sum_{\mu=1}^{4} z^\mu A_{j,\mu}(z) = 0 \qquad (z \in \mathbb{R}^4)\,. \tag{6.6}$$

A real jet is a sequence $J = (A_j)_{j\ge 0}$ with $A_j \in \mathcal{A}_j$. Choose real coordinates on each $\mathcal{A}_j$, assigning weight $j+2$ to each coordinate. Let V_d be the space of complex polynomials in these coordinates, of total weight at most d, invariant under simultaneous conjugation by G.

The coefficients in Definition 6.3 are Taylor coefficients, with the factorial normalization, rather than derivative values. Equivalently, they are diagonal evaluations of symmetric $(j+1)$-linear maps.

The weight $j+2$ records the scaling of a link integral: the polynomial one-form contributes $j+1$ powers of length and integration contributes one more. Weights add under multiplication. We also count the number of factors in a monomial, its ordinary polynomial degree; the two gradings will serve different purposes.

An orthogonal matrix h acts on these one-forms and on their invariant polynomials by

$$(T_h A_j)_\mu(z) = \sum_\nu h_{\mu\nu} A_{j,\nu}(h^{-1}z), \qquad (h \cdot v)(J) = v(T_{h^{-1}} J)\,. \tag{6.7}$$

This action preserves (6.6) and commutes with conjugation by G. Coefficient conjugation is denoted by $v \mapsto \bar{v}$; a real polynomial has $\bar{v} = v$. Let $V_d^{w,f}$ be the subspace homogeneous of weight w and factor number f.

The weight bound leaves only finitely many coordinates and monomials. We can therefore choose one finite basis at each degree, retaining the basis elements already chosen at lower degrees. Real homogeneous bases make conjugation and the two gradings explicit.

Lemma 6.4 (Finite label spaces)**.** *For the spaces of Definition 6.3, evaluation on real jets is injective, and*

$$\dim_{\mathbb{C}} V_d < \infty, \qquad 1 \in V_d \subseteq V_{d+1}, \qquad V_d = \bigoplus_{\substack{0 \le w \le d \\ 0 \le f \le w}} V_d^{w,f}\,. \tag{6.8}$$

Each summand is preserved by (6.7) and coefficient conjugation. There are nested real bases, homogeneous in both gradings and containing 1*.*

Proof. Only coordinates with $j+2 \le d$ can occur in a monomial of weight at most d. There are finitely many such coordinates, and the exponent of each is bounded by d. Thus finitely many monomials span the polynomials under consideration, and their invariant subspace is finite dimensional. A complex polynomial vanishing on every real jet vanishes identically, by the one-variable polynomial identity theorem applied successively in these finitely many real coordinates.

Conjugation by G acts linearly within each $\mathcal{A}_j$. It preserves weight and factor number, so comparison of homogeneous components of the invariance identity shows that each component remains invariant. Every coordinate has weight at least two; in particular the factor number cannot exceed the weight, giving the index range in (6.8). The same linearity argument applies to (6.7), and these real linear changes commute with coefficient conjugation. Each invariant homogeneous component is the complex span of its real polynomials, by decomposition into real and imaginary coefficient parts. Choose a real basis in each component, beginning with $\{1\}$ for the constant component, and retain these choices as d increases. □

Fix the bases of Lemma 6.4, writing $b_{d,i}$ for their elements and $(w_{d,i}, f_{d,i})$ for their two grades, with $b_{d,0} = 1$. We identify each polynomial with its coefficient vector when applying a linear map to V_d. The continuum spaces F_d of Definition 2.17 will arise only after a limiting correlation table and its null relations have been obtained. The spaces V_d still contain all classical labels, before these relations are imposed.

6.3 Local representatives

A polynomial in Taylor coefficients is not yet a lattice observable. We seek a polynomial in nearby link matrices whose Taylor expansion, when tested on a classical connection, recovers the chosen label. This is the purpose of a local section. Its existence is a separate construction; here we state its normalization precisely.

On $\mathbb{Z}^4$, the gauge action of Definition 2.5 and the coordinate pullbacks of Definition 6.1 use the same formulas as on a torus. Polynomial functions on a finite set of links can then be evaluated on any periodic lattice.

Definition 6.5 (Local polynomial functions)**.** For a finite set $\Sigma \subset \mathbb{Z}^4$, let $\mathcal{P}(\Sigma)$ be the polynomial functions of matrix entries and conjugate entries of links with both endpoints in Σ. At $o \in \Lambda_\ell$, evaluate such a polynomial by substituting

$$U(x,\mu) \longmapsto U(o + [x]_\ell, \mu)\,. \tag{6.9}$$

The link conventions are those of Definition 2.2.

This implements the finite-site locality requirement Reg4 of Definition 2.8. The substitution is meaningful even when distinct cover links become the same torus link.

To test the classical normalization, truncate a real jet of Definition 6.3 to weights at most d and set $A_\mu^J = \sum_{j+2\le d} A_{j,\mu}$. For a spacing parameter $\varepsilon \in \mathbb{R}$, take the transport

$$\frac{d}{dt}u(t) = u(t)\varepsilon A_\mu^J(\varepsilon(x + te_\mu)), \qquad u(0) = 1, \qquad U_{J,\varepsilon}(x,\mu) = u(1)\,. \tag{6.10}$$

Successive integration gives a normally convergent series on compact parameter sets: the term with r integrations is bounded by $\frac{M^r}{r!}$ if the coefficient matrix is bounded by M on $[0,1]$. The series solves the equation, and iteration of the integral equation proves uniqueness. In particular, the endpoint is real analytic in ε near 0, for each fixed real jet. Approximating the coefficient by step functions gives convergent products of exponentials from $\mathfrak{g}$; their limits lie in the closed group G. Thus (6.10) gives actual group-valued links, with the transport convention of Definition 2.2.

The section must preserve the exact lattice symmetries and the constant observable. Its classical normalization requires equality of Taylor coefficients through the degree being represented. These are the conditions in the next definition.

Definition 6.6 (Normalized local section)**.** A normalized local section in degree d is a finite set $\Sigma_d \subset \mathbb{Z}^4$ and a complex linear map $P_d : V_d \to \mathcal{P}(\Sigma_d)$, with polynomial functions as in Definition 6.5, such that

$$0 \in \Sigma_d, \qquad h\Sigma_d = \Sigma_d \quad (h \in H_4), \tag{6.11}$$

and, for all $v \in V_d$, link configurations U on $\mathbb{Z}^4$, $\gamma : \mathbb{Z}^4 \to G$ and $h \in H_4$,

$$\begin{aligned} P_d[v](U^\gamma) &= P_d[v](U), & P_d[1] &= 1, \\ P_d[\bar v](U) &= \overline{P_d[v](U)}, & P_d[v](h^*U) &= P_d[h\cdot v](U)\,. \end{aligned} \tag{6.12}$$

For every real jet J and $0 \le k \le d$, it also satisfies

$$\frac{d^k}{d\varepsilon^k} P_d[v](U_{J,\varepsilon})\Big|_{\varepsilon=0} = \frac{d^k}{d\varepsilon^k} v(\mathrm{sc}_\varepsilon J)\Big|_{\varepsilon=0}, \qquad (\mathrm{sc}_\varepsilon J)_j = \varepsilon^{j+2} A_j\,. \tag{6.13}$$

Here V_d, its real jets and its action are as in Definition 6.3 and (6.7); H_4 and its pullbacks are from Definition 6.1, and the test links are (6.10).

The Taylor identity imposes no equality at nonzero spacing. No compatibility of P_d and P_{d+1} is required. The existence proof must also give derivative estimates for the later blocking argument; Definition 6.6 does not assert those estimates.

6.4 Renormalized insertions

As shorter scales are integrated out, the coefficients of an observable change and different classical labels can mix. We normalize the observable by its coefficients at one reference level $r(d)$, chosen independently of cutoff and volume. The inverse of the coefficient transfer then determines which fine-lattice polynomial represents a prescribed reference label. This describes a particular choice of the insertions in Definition 2.6, not an additional condition on that definition.

Use a_N, N, p from Definition 2.7, the basis from Lemma 6.4, and a local section from Definition 6.6. Let $\theta_N > 0$ be the threshold that will distinguish small plaquette distances χ in the later estimates. Define the linear factor scaling by $S_t b_{d,i} = t^{f_{d,i}} b_{d,i}$. For an invertible complex linear map $\tau_{N,p,d} : V_d \to V_d$, the candidate insertion is

$$I_{N,p}((d,v),o;U) = \frac{1}{a_N^d} P_d\Big[S_{\theta_N^{-1}} \tau_{N,p,d}^{-1} v\Big]\,(o;U), \tag{6.14}$$

with placement (6.9). Thus a label with f factors is multiplied by $\frac{1}{\theta_N^f}$; the prefactor $\frac{1}{a_N^d}$ supplies the physical degree normalization.

For $N \geq r(d)$, let $x_j : V_d \to V_d$ be complex linear maps from coefficients at the finest level N to coefficients at level j, for $r(d) \leq j \leq N$. The conditions that make $\tau_{N,p,d}$ a reference transfer are

$$x_N(v) = v, \qquad x_{r(d)}(v) = \tau_{N,p,d} v, \qquad x_j(a_N^d 1) = a_j^d 1 \quad (r(d) \leq j \leq N)\,. \tag{6.15}$$

These equalities fix how coefficients are normalized between levels; they assert no pointwise identity between observables on different lattices. Reality and signed-permutation covariance require

$$\tau_{N,p,d}\bar{v} = \overline{\tau_{N,p,d} v}, \qquad \tau_{N,p,d}(h \cdot v) = h \cdot (\tau_{N,p,d} v) \quad (h \in H_4)\,. \tag{6.16}$$

The locality of the resulting insertion follows from its polynomial formula, even on a torus smaller than the chosen set of sites. The unit normalization is different: it uses the reference-transfer conditions and therefore applies only beyond the reference level. The following proposition separates these two conclusions.

Proposition 6.7 (Locality of the insertions)**.** *Let P_d satisfy Definition 6.6, $\theta_N > 0$ and $\tau_{N,p,d} \in \mathrm{GL}_{\mathbb{C}}(V_d)$. For all N, p, the map $v \mapsto I_{N,p}((d,v),o;\cdot)$ in (6.14) is complex linear and satisfies Reg2–Reg4 of Definition 2.8, with finite set Σ_d. Moreover,*

$$I_{N,p}((d,v),o;U(\cdot + z)) = I_{N,p}((d,v),o+z;U)\,. \tag{6.17}$$

If $N \geq r(d)$ and (6.15)–(6.16) hold, then

$$\begin{aligned} I_{N,p}((d,\bar{v}),o;U) &= \overline{I_{N,p}((d,v),o;U)}, \\ I_{N,p}((d,h\cdot v),A(o);U) &= I_{N,p}((d,v),o;A^*U), \\ I_{N,p}((d,a_{r(d)}^d 1),o;U) &= 1, \qquad x_{r(d)}(\tau_{N,p,d}^{-1} v) = v \end{aligned} \tag{6.18}$$

for $A(o) = ho + z$, $h \in H_4$, as in Definition 6.1.

Proof. The coefficient changes in (6.14) are complex linear. A polynomial in link entries and their conjugates is continuous, and the section is gauge invariant by (6.12); this proves Reg2 and Reg3. Placement (6.9) reads only links with both endpoints in $o + [\Sigma_d]_{L_p D_N}$, proving Reg4. Identifying distinct cover vertices does not enlarge that set. Translating the placement proves (6.17).

Factor scaling commutes with coefficient conjugation and H_4, since both actions preserve factor number by Lemma 6.4. Together with (6.16) and (6.12), this gives the first two identities in (6.18). The endpoint and unit conditions (6.15) yield $\tau_{N,p,d}(a_N^d 1) = a_{r(d)}^d 1$. Since $S_t 1 = 1$ and $P_d[1] = 1$, substitution in (6.14) gives the unit identity. The remaining identity follows by applying the reference endpoint condition to $\tau_{N,p,d}^{-1} v$. □

Choose the lattice observable labels in Definition 2.6 to be $\mathcal{T} = \bigsqcup_d V_d$. Under the hypotheses of Proposition 6.7, the requirements Reg2–Reg4 are satisfied at every cutoff; the reference unit is $a_{r(d)}^d 1$. Proposition 6.2 makes every lattice moment in (2.9) an integral of a continuous, hence integrable, product of observables. The running-coupling condition Reg1 and convergence of these moments require the later ultraviolet arguments.

Unresolved. Two construction steps are still missing from the manuscript: producing the normalized polynomial sections of Definition 6.6 with the derivative bounds needed for blocking, and producing invertible coefficient transfers satisfying (6.15)–(6.16) for the same Wilson family. Proposition 6.7 assumes these inputs. Their source locations are recorded below; their paper proofs belong to later ultraviolet sections.[6]

7 Infinite volume

A correlation function records the joint expectation of finitely many local observables. We seek its limit as the torus grows, using the moments of Definition 2.9 for the Wilson family of Definition 2.7. The estimates must hold independently of the torus size, so that the same subsequence works for every observable and every separated test of Definition 2.10.

There are two steps. Reflection positivity gives bounds and continuity estimates, from which we extract limiting correlations. To differentiate those limits, we must also identify which lattice observables represent derivatives and which disappear from every correlation. That identification needs the equation comparisons and a finite-dimensional rank estimate stated below. We initially refine the spacing sequence as well as the volume sequence; the full fixed-volume limit of Proposition 5.1 removes the spacing refinement at the end. The phase decomposition in Proposition 5.2 requires later arguments and is not part of this section.

7.1 Lattice estimates

We use the polynomial coefficient spaces V_d of Lemma 6.4 and the insertions $I_{N,p}$ of (6.14). Use the fixed bases there to choose norms on V_d and operator norms on maps between these spaces. A list $v = (v_1, \dots, v_n)$ always includes specified degrees d_i, with $v_i \in V_{d_i}$; we suppress these degrees in $I_{N,p}(v_i, y)$. The action (6.7), restricted to the signed coordinate permutations $h \in H_4$, is written $\rho_d(h)$. Coefficient conjugation and the unit $u_d = a_{r(d)}^d 1$ are those of Proposition 6.7; in particular, the reference-normalized unit need not be the polynomial 1.

The following scale conditions are specific to the constructed family, beyond Definition 2.7:

$$a_0 = 1, \qquad a_{j+1} = \frac{a_j}{\ell_j}, \qquad \ell_j \in 2\mathbb{N}+1, \quad \ell_j \geq 3, \qquad L_p \in 4\mathbb{N}, \qquad \beta_{N,p} \geq 0\,. \tag{7.1}$$

[6] Formal correspondence at revision `3456e5e`: Local sections: `YM/Integrated.lean`, lines 214–219. Transfer conditions: `YM/Source/Insertion.lean`, lines 89–114; same-family application: `YM/FullVolume/Cross.lean`, lines 30–71.

Thus one step at scale a_j comprises ℓ_j steps at scale a_{j+1}. We also require these block factors to grow slowly relative to the inverse spacing:

$$\frac{\log \ell_j}{\log a_j^{-1}} \longrightarrow 0 \qquad (j \longrightarrow \infty)\,. \tag{7.2}$$

The quotient at $j = 0$ plays no role. The translation, conjugation, signed-permutation and unit identities in Proposition 6.7 are assumed beyond the reference levels.

To use reflection positivity, divide the torus into congruent cubical cells, with the same even number of cells in each coordinate direction. In each tiling, all dividing planes pass through lattice sites, or all bisect links; these give site and bond reflections, respectively. An observable is interior to a cell if it is bounded, measurable, gauge invariant and depends only on links with both endpoints strictly inside that cell. For such an observable F in the base cell of a tiling $\mathcal{C}$, reflect it into every cell. Conjugate its value whenever an odd number of coordinate reflections has been used, and denote the resulting copy in cell C by $\Theta_C F$. The chessboard quantity compares one local observable with a product of its reflected copies filling the torus:

$$\|F\|_{\mathcal{C}} = \left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F \right\rangle_{N,p} \right|^{\frac{1}{|\mathcal{C}|}} . \tag{7.3}$$

The centered tiling $\mathcal{C}_j(y)$ has physical cell side a_j and base cell centered at y. The ultraviolet estimate bounds this quantity at every fixed physical cell size. For every $d \in \mathbb{N}$ and $R, \varepsilon > 0$, there exists $A \geq 0$ such that, for every $j \geq r(d)$, eventually in N, uniformly in p, y and $\|v\|_{V_d} \leq R$, the insertion is interior and

$$\|I_{N,p}(v, y)\|_{\mathcal{C}_j(y)} \leq A a_j^{-d-\varepsilon}\,. \tag{7.4}$$

The cutoff threshold may depend on j. A fixed finite collection of transformed labels is included by enlarging R. [7]

Unresolved. The ultraviolet part must establish (7.1), (7.2), (7.4) and the insertion identities for the particular family of Proposition 5.1. A bound for a different family would not suffice. These are the hypotheses used below, not consequences of Definition 2.8.[8]

Reflection positivity is the inequality that makes the chessboard bound possible. If r reflects the lattice in one of the dividing planes, write $\Theta_r F = \overline{F \circ r^*}$, where r^* reflects the link configuration. It pairs an observable on one side of the plane with its reflected complex conjugate on the other side.

Lemma 7.1 (Reflection estimates)**.** *For the Wilson measure of Definition 2.4 on an even torus, $\beta \geq 0$, and observables interior to the indicated half-torus or cells,*

$$\langle \Theta_r F\, F \rangle_\beta \geq 0 \qquad \textit{(r a site or bond reflection)}, \tag{7.5}$$

$$\left| \left\langle \prod_{C \in \mathcal{C}} \Theta_C F_C \right\rangle_\beta \right| \leq \prod_{C \in \mathcal{C}} \|F_C\|_{\mathcal{C}}, \tag{7.6}$$

$$\|F + G\|_{\mathcal{C}} \leq \|F\|_{\mathcal{C}} + \|G\|_{\mathcal{C}}.$$

Here the chessboard quantities are those of (7.3).

[7] Formal correspondence at revision `3456e5e`: `Extraction/Center.lean`: the uniform centered estimate and its projection to each extraction.

[8] Formal correspondence at revision `3456e5e`: `MergeProof/Gardens.lean` and `Source/Construction.lean`.

Proof. The kernel $(U, V) \mapsto \operatorname{Re}\operatorname{tr}(UV^*)$ on G is positive semidefinite: it is the real Euclidean inner product of the matrix entries. Its exponential with a nonnegative coefficient is positive semidefinite by the power series and Schur products. Multiplication by the two half-slice weights preserves positivity. Let X, Y be configurations on the two boundary slices. After fixing the crossing links to the identity, their interaction and half-slice weights define a positive kernel $K_0(X, Y)$. Haar invariance restores the crossing-link integration as a double average over the boundary gauge variables g, h:

$$K(X, Y) = \iint K_0(g \cdot X, h \cdot Y)\, dg\, dh\,.$$

Its quadratic forms are those of K_0 integrated against finite combinations of these averaged point masses, hence are nonnegative. The use of independent variables in the two arguments is essential. Splitting a Wilson configuration into the fixed planes, crossing links and open halves expresses its reflected form as an integral of these positive boundary forms. For a site seam the boundary form is a squared modulus. The change of variables preserves product Haar measure, and division by the positive partition function preserves positivity. All integrands are bounded on compact configuration spaces. [9]

A fold replaces the observables on one half of the torus by reflected copies of those on the other half. Cauchy–Schwarz bounds the square of the absolute expectation of a mixed cell assignment by the product of its two folded assignments. The finite folding principle on such a periodic tiling says that a nonempty class closed under both folds contains a constant assignment. To see this, fix a coordinate and start with a band of width one cell, on which every assignment is independent of that coordinate. Folding at its initial seam copies its first layer to the preceding layer and enlarges the band by one cell, as long as its width does not exceed half the period. Once the band fills a half-period, one more fold makes the assignment independent of that coordinate everywhere. Each fold preserves independence of the coordinates already treated. Repeat in the remaining coordinates. Apply this principle to maximizing assignments from any fixed finite list of functions. If their maximum is $M > 0$, the fold inequality $M^2 \le M_1 M_2$, with $0 \le M_i \le M$, keeps both folds maximal; if $M = 0$, every assignment is maximal. Thus a constant assignment attains the maximum. Normalize each F_C by $\|F_C\|_{\mathcal{C}} + \eta$ and let $\eta \downarrow 0$ to obtain (7.6). Finally expand the reflected product of $F + G$, apply (7.6) to every term, and take the $|\mathcal{C}|$th root. [10] □

7.2 Bounds and compactness

Centered estimates control an insertion well inside a translated larger cell. For aligned fine-cell centers, repeat the larger-cell observable throughout the torus and apply (7.6) on the finer tiling, putting 1 in unused cells. The root is bounded by the largest centered root of the transformed labels. In one coordinate, let $R(z)$ be the chessboard quantity as the anchor moves by z lattice steps inside a fixed cell. Between consecutive aligned centers, translation invariance and the fold estimate give

$$R(z)^2 \le R(z-1)R(z+1)\,.$$

A nonnegative log-convex sequence is bounded by its endpoint maximum. Applying this in each coordinate proves the translated-cell bound when the anchor is at least $3a_k$ from the boundary and

[9] Formal correspondence at revision `3456e5e`: `Slice/Positive.lean`: positivity of the weight and double average; `Slice/Reflection.lean`: reflected positivity.

[10] Formal correspondence at revision `3456e5e`: `Cells/Flood.lean`: the folding principle; `Cells/Chess.lean`: maximization, the product estimate and subadditivity.

the links on which it depends lie within distance a_k, where a_k is the finer cell side. [11]

Suppose n insertion sites have mutual physical torus sup-distance at least r. Choose $m \le k$ with $a_m < r$ and $6na_k < a_m$. In each coordinate the translations placing an anchor within $3a_k$ of a cell boundary occupy at most $6na_k$ of one period a_m. Choose a translation outside these strips. Different sites occupy different cells, so the translated-cell bound and (7.6) bound their correlation by A^n. Here A bounds the finitely many centered roots at level k, and the finite sets of links fit once N is sufficiently large. This gives local bounds at each fixed volume and bounds independent of the volume on every compact subset of Conf_n along ordered tails. An ordered tail means: first the volume index is large, then the spacing index is large with a threshold allowed to depend on that volume.

The transfer argument separates insertions by coordinate hyperplanes. We therefore first work on configurations whose points have different projections onto each coordinate axis:

$$\mathrm{Gen}_n = \{x \in \mathrm{Conf}_n : x_i^\mu \neq x_j^\mu \text{ for } i \neq j,\ \mu = 1, \dots, 4\}\,. \tag{7.7}$$

At fixed period replace inequality of coordinates by inequality modulo $L_p\mathbb{Z}$, and denote the set by $\mathrm{Gen}_n^{(p)}$; it is contained in the periodically separated set of Definition 2.13. These sets are open and dense in the distinct-point configurations (2.10); their complements have measure zero. For example, two distinct points with the same first coordinate lie outside Gen_2. Smoothness will later recover the correlations at such configurations.

Equicontinuity on these sets comes from propagation between parallel lattice slices. Fix a coordinate direction, let M be the number of slices around the torus, and enclose each insertion in a slab of w consecutive slices. Integration over the links between successive slices gives a positive kernel K. Let B_i be the kernel obtained by integrating within the ith slab with its observable inserted, and let Y_i be the number of steps between slabs. Assigning half the slice action to each neighbor expresses the correlation as

$$\frac{\mathrm{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})}{Z}, \qquad Z = \mathrm{Tr}\, K^M > 0, \qquad nw + \sum_i Y_i = M\,.$$

Here traces denote cyclic integrals of kernels on the compact slice configuration space. The kernels are continuous. If A_i bounds the chessboard quantity of the ith observable and $M = 2qw$, the reflected-product bound is

$$\left(\frac{\mathrm{Tr}((B_i^* B_i)^q)}{Z}\right)^{\frac{1}{2q}} \le A_i\,.$$

Apply (7.6) to $2q$ alternating copies of the observable and its reflected conjugate. Divisibility of the period by four supplies the integer q when the slabs have twice a centered cell's width.

Changing two distances between slabs by $+h$ and $-h$, where $0 \le h \le b$ and all original distances exceed $2b$, changes the normalized trace by at most $(\prod_i A_i)^{\frac{h}{b}}$. Here is the estimate that avoids a factor depending on the dimension of the slice space. Approximate every cyclic integral by the same finite weighted sum, with nonnegative weights. For its positive matrix K, define $K^z = \exp(z \log K)$ on the positive eigenspaces and zero on the kernel. The trace with two exponents $Y_i + z, Y_j - z$ is entire. For a matrix T, write $\|T\|_s = (\mathrm{Tr}\,|T|^s)^{\frac{1}{s}}$ for its Schatten norm. For $|z| \le 2b$, the Schatten Hölder inequality bounds the trace by

$$\prod_i \|B_i\|_{2q}\, Z^{1-\frac{n}{2q}}, \qquad \frac{n}{2q} + \frac{\sum_i \mathrm{Re}\, Y_i(z)}{M} = 1\,.$$

[11] Formal correspondence at revision `3456e5e`: `Extraction/Around.lean`: transfer to interior positions; `Cells/Slide.lean`: translation log-convexity.

Imaginary powers are contractions, and the real powers have the indicated Schatten norms. Cauchy's estimate on discs of radius b centered on $[0,h]$ bounds the derivative by this expression divided by b. Integrate and pass to the limit in the finite weighted sums. Every endpoint exponent is positive, so the convention on the kernel changes no endpoint value. [12]

A site move changes two distances. In physical units the preceding estimate gives a Lipschitz bound; telescoping over the $4n$ coordinates gives equicontinuity on compact subsets of (7.7). The centered estimates make its constant independent of the volume along ordered tails. For $n=1$ use translation invariance; for $n=0$ the moment is 1. No derivative bound has been used.

These bounds let us choose one subsequence for all correlations at once. The essential point is that each coefficient space has a finite basis, while the arities, degrees, volumes and compact exhaustions form countable collections. Write $M^{\mathrm{step}}_{N,p}(v;x)$ for the moment (2.9) evaluated at the grid configuration obtained by rounding x to the lattice.

Proposition 7.2 (Auxiliary extraction)**.** *Assume* (7.1), (7.2) *and* (7.4) *and the insertion identities of Proposition 6.7. Every pair of strictly increasing spacing and volume sequences has refinements N_j, p_k and multilinear kernels P_p, S_0 with the following properties, simultaneously for all $n \in \mathbb{N}$, degrees $d_i \in \mathbb{N}$ and coefficients $v_i \in V_{d_i}$.*

(i) *Locally uniformly on the sets* (7.7),

$$\begin{aligned} M^{\mathrm{step}}_{N_j,p}(v;\cdot) &\longrightarrow P_p(v;\cdot) \in C(\mathrm{Gen}_n^{(p)}) \quad (p \in \mathbb{N}),\\ P_{p_k}(v;\cdot) &\longrightarrow S_0(v;\cdot) \in C(\mathrm{Gen}_n). \end{aligned} \tag{7.8}$$

(ii) *Every separated test f of Definition 2.10 is eventually supported inside volume p_k (Definition 2.11), has $fS_0(v;\cdot) \in L^1$, and admits numbers $R_p(v;f)$ such that*

$$\begin{aligned} &\lim_{j\to\infty} C_{N_j,p}(v;f) = R_p(v;f) \qquad\qquad (f \text{ supported inside volume } p),\\ &\lim_{k\to\infty} R_{p_k}(v;f) = \int f(x) S_0(v;x)\,dx. \end{aligned} \tag{7.9}$$

Proof. Extend the lattice functions by rounding points to the grid. Rounding moves a configuration by $O(na_N)$; on a compact neighborhood inside (7.7) the oscillation of these step functions is therefore bounded by $C|x-y| + o(1)$, uniformly in x, y. Choose a countable dense set in each such neighborhood and extract convergence at its points. A finite net and this oscillation estimate make the subsequence uniformly Cauchy on each compact set; its limit is continuous. This is the compactness argument underlying Arzelà–Ascoli, without requiring the step functions themselves to be continuous. Apply it over all fixed periods, arities, degree lists, basis labels and compact exhaustions of (7.7). These indices are countable, so one spacing subsequence gives all the kernels P_p. The volume bounds and moduli pass to P_p by taking the spacing limit at moving lattice points. A second countable extraction gives the volume subsequence and S_0. Extend from basis labels by multilinearity. Set the raw kernels equal to zero off their coordinate-separated sets (7.7); these measurable extensions change no integral.

Write each weighted lattice sum as an integral of a step function. A compact neighborhood of $\operatorname{supp} f$ remains separated; at a fixed volume containing its support in Definition 2.11, it also avoids periodic collisions. Local bounds dominate the integrands there, and rounding changes f

[12] Formal correspondence at revision `3456e5e`: `Complex/Gaps.lean`: trace bounds and finite differences; `Slice/Calendar.lean`: kernel passage; `Extraction/Difference.lean`: columns and one-coordinate shifts.

uniformly by a quantity tending to zero. Almost-everywhere convergence on $\mathrm{Gen}_n^{(p)}$ gives the first limit by dominated convergence. Apply dominated convergence again on $\operatorname{supp} f$, using the volume-independent bound and the full-measure set Gen_n. This gives the second limit and integrability. Eventual containment in the volume follows from $L_{p_k} \to \infty$. [13] □

7.3 Equation relations

Compactness gives correlations only where all coordinate projections are distinct. To extend them smoothly across the remaining configurations, we identify every weak derivative with the correlation of another observable. A lattice difference is represented by a label of higher degree, whose coefficients may grow as the spacing shrinks. The equation comparisons remove the part invisible to correlations; the rank estimate prevents the remaining coefficients from escaping to infinity.

We now state these additional ultraviolet inputs precisely. Along a further common refinement, preserving Proposition 7.2, assume linear projections $P_{d,N,p}, P_d : V_d \to V_d$ with

$$P_{d,N,p}^2 = P_{d,N,p}, \qquad P_{d,N,p} \longrightarrow P_d, \qquad P_d^2 = P_d \quad \text{in operator norm.} \tag{7.10}$$

All limits and eventual assertions in this subsection use the ordered tails described in Section 7.2. The inverse reference map in (6.15) measures the size of a coefficient family $w_{N,p}$ at the fine cutoff. We allow families satisfying

$$\forall \varepsilon > 0, \qquad \|\tau_{N,p,d}^{-1} w_{N,p}\|_{V_d} \le a_N^{-\varepsilon} \quad \text{eventually.} \tag{7.11}$$

The ultraviolet estimates must imply (7.11) for bounded coefficient families. They do so by bounding the growth at each intervening scale: the logarithm of the product of these bounds is $o(\log a_N^{-1})$.

An error $E_{N,p}(y)$ is small in reflected cells if one nonnegative rate $\eta_{N,p} \to 0$ along ordered tails has the following property. For every integer $q \ge 0$ and lower level m, there exists $k \ge m$ such that eventually, for every translated bond cell of side a_k and every anchor with relative lattice coordinates z satisfying

$$M \le z^\mu, \quad z^\mu + M + 1 \le \frac{a_k}{a_N}, \quad M = \left\lfloor \frac{a_k}{8(q+1)a_N} \right\rfloor \quad (\mu = 1, \dots, 4), \tag{7.12}$$

the error is interior and its root is at most $\eta_{N,p}$. This condition keeps the error away from cell boundaries, where the chessboard estimate can be applied. A degree-e representative means an insertion $I_{N,p}(w_{N,p}, y)$ with $w_{N,p} \in V_e$. At every degree we require the following three comparisons:

(C1) Each fixed degree-d insertion has a degree-e representative satisfying (7.11) for $d \le e$, differing by an error small in reflected cells.

(C2) The forward difference $a_N^{-1}(I(v, y + e_\mu) - I(v, y))$ has such a representative of degree $d + 1$.

(C3) A family satisfying (7.11) in $\ker P_{d,N,p}$ differs by such an error from bounded measurable observables whose correlations with every fixed list of the other insertions vanish exactly, eventually at each positive separation.

[13] Formal correspondence at revision `3456e5e`: `Extraction/Lipschitz.lean`: the two extractions; `Extraction/Integrate.lean`: dominated convergence; `Extraction/Held.lean`: their combination.

Boundedness in C3 is only at each cutoff and site, which suffices for integration against the finite Wilson law.

Fix an arity, one slot i, all coefficients in the other slots, and a separated test f of Definition 2.10. Varying the coefficient $w \in V_d$ in that slot gives linear functionals

$$L_{N,p}(w) = C_{N,p}(v_{i\to w}; f), \qquad L(w) = \int f(x) S_0(v_{i\to w}; x)\, dx, \qquad N_d = \bigcap_L \ker L. \tag{7.13}$$

Here $v_{i\to w}$ means replacement of the coefficient in slot i by w; the intersection includes every choice of the other data. Thus N_d consists of coefficients invisible to every separated smeared correlation. The additional rank input is

$$\dim_{\mathbb{C}} N_d \le \dim_{\mathbb{C}} \ker P_d\,. \tag{7.14}$$

The projectors and this inequality are obtained on a further common refinement, retaining all previous limits.

Unresolved. Supply comparisons C1–C3, the coefficient bound (7.11) for bounded families, and the finite-dimensional detection argument for (7.14) for the chosen family. The dimension inequality is additional to the centered-root estimate.[14]

The following lemma replaces the cutoff-dependent coefficients in these comparisons by fixed coefficients. Its conclusion holds against every list of other fields, so that it gives genuine field operations rather than representatives depending on a particular test.

Lemma 7.3 (Fixed representatives)**.** *Assume the hypotheses of Proposition 7.2, the projections (7.10), the coefficient bound (7.11) for bounded families, the error condition (7.12), the rank bound (7.14), and comparisons C1–C3. Then $N_d = \ker P_d$, and there are complex-linear maps*

$$\kappa_{d,e} : V_d \longrightarrow V_e \quad (d \le e), \qquad \delta_{d,\mu} : V_d \longrightarrow V_{d+1}$$

such that, for every separated test of Definition 2.10 and functionals (7.13),

$$\begin{aligned} \int f S_0(v_{i\to \kappa_{d,e} v_i}) &= \int f S_0(v), \\ \int f S_0(v_{i\to \delta_{d,\mu} v_i}) &= -\int (\partial_{i,\mu} f) S_0(v). \end{aligned} \tag{7.15}$$

Proof. An error small in reflected cells has vanishing separated smearings. Choose its cell below the fixed separation and a finer centered scale for the other insertions. Avoid boundary strips as before, with margin M; the factor $8(q+1)$ permits this for any fixed number of slots. The translated-cell estimate and (7.6) bound the mixed moment by $C\eta_{N,p}$. The weighted count of sites meeting $\operatorname{supp} f$ is bounded by the volume of a fixed neighborhood, independently of N, p. Thus its smear tends to zero.

The maps $L_{N,p}$ converge in operator norm because they converge on a finite basis. For fixed v, the bounded family $v - P_{d,N,p} v$ belongs to the projection kernel. Comparison C3 gives $L(v - P_d v) = 0$ for every L. Thus $\ker P_d \subset N_d$, and (7.14) gives equality.

Summation by parts on the torus gives

$$a_N^{4n} \sum_y f(\widehat{y}) \Delta_{i,\mu} M_{N,p}(v; y) = a_N^{4n} \sum_y \frac{f(\widehat{y} - a_N e_{i,\mu}) - f(\widehat{y})}{a_N} M_{N,p}(v; y)\,.$$

[14] Formal correspondence at revision `3456e5e`: `MergeProof/Powder.lean` and `Next/RANK/Beam.lean`: same-family assembly and retained rank comparison.

Here $\Delta_{i,\mu}$ is the forward quotient in slot i, and $e_{i,\mu}$ its unit coordinate vector. The test support stays inside the fundamental box eventually, so there is no wrapping. Its backward quotient tends uniformly to $-\partial_{i,\mu} f$ on a fixed neighborhood. Local boundedness and the bounded weighted site count make the error vanish. Comparisons C1–C2 therefore give coefficient families $w_{N,p}$ with the required target limits for every choice of the other insertions and test at once.

Write $p_{N,p} = P_{e,N,p} w_{N,p}$, where e is the representing degree. Comparison C3 applied to the normalized remainder gives

$$L_{N,p}\left(\frac{w_{N,p} - p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right) \longrightarrow 0\,.$$

Division by $1 + \|p_{N,p}\|_{V_e} \geq 1$ preserves (7.11), and the normalized projected vector is bounded. Take a cluster limit (c, z) of

$$\left(\frac{1}{1 + \|p_{N,p}\|_{V_e}}, \frac{p_{N,p}}{1 + \|p_{N,p}\|_{V_e}}\right)$$

in $[0,1] \times \overline{B}_{V_e}(0,1)$, using a refining ultrafilter of the ordered tails. Then $P_e z = z$, $c + \|z\|_{V_e} = 1$, and $Lz = c\, t_L$ for every target t_L. If $c = 0$, every L annihilates z, so $z \in N_e = \ker P_e$ and $z = 0$, a contradiction. Thus $\frac{z}{c}$ represents every target simultaneously. Choose representatives on each source basis and extend linearly. No further subsequence has been chosen for the uncountable collection of tests. [15] □

7.4 Smooth fields and positivity

Lemma 7.3 represents each weak derivative by another correlation function. Repeating this operation gives weak derivatives of every order, and hence a smooth representative even where coordinate projections coincide. A coefficient vanishing in every correlation is discarded; the resulting quotient is the space of continuum fields of that degree. The maps between degrees and the derivatives are independent of all choices of coefficient representatives. These will be the finite-degree spaces and projection in Definition 2.17.

Proposition 7.4 (Smooth correlations)**.** *Under the hypotheses of Lemma 7.3, there are multilinear kernels S_n and quotient maps $\pi_d : V_d \to F_d = V_d/N_d$ such that:*

(i) *$S_n(v;\cdot) \in C^\infty(\mathrm{Conf}_n)$ on (2.10), $S_n|_{\mathrm{Gen}_n} = S_0$ on (7.7), and (7.9) holds with S_n.*

(ii) *$\ker \pi_d = N_d = \ker P_d$ is the null space of S: a coefficient lies in N_d exactly when every separated-point correlation containing it vanishes.*

(iii) *The maps $\kappa_{d,e}, \delta_{d,\mu}$ of Lemma 7.3 induce linear maps $j_{d,e} : F_d \hookrightarrow F_e$ and $\partial_\mu : F_d \to F_{d+1}$, with $j_{e,f} j_{d,e} = j_{d,f}$ and $j_{d,d} = 1$, satisfying*

$$\begin{aligned} S_n(\dots, j_{d,e} A_i, \dots; x) &= S_n(\dots, A_i, \dots; x), \\ S_n(\dots, \partial_\mu A_i, \dots; x) &= \partial_{i,\mu} S_n(\dots, A_i, \dots; x). \end{aligned} \tag{7.16}$$

The notation $S_n(A; x)$ denotes the induced correlation of quotient fields, with $A_i = \pi_{d_i} v_i$.

Proof. Equation (7.15) represents every weak coordinate derivative by another locally integrable kernel. Iteration gives derivatives of every order. For a smooth cutoff χ supported in Conf_n, each

[15] Formal correspondence at revision `3456e5e`: `Work/D01.lean`–`D04.lean`: coefficient costs, errors and differences; `Dash/Limits.lean`: normalized compactness; `Dash/Inert.lean`: fixed representatives.

derivative of $\chi S_0(v)$ is a finite sum of integrable compactly supported kernels. Integration by parts in the Fourier transform gives, for every integer $b \geq 0$,

$$\sup_{\xi}(1+|\xi|)^b \left|\widehat{\chi S_0(v)}(\xi)\right| < \infty\,.$$

Fourier inversion supplies a smooth representative. Choose $\chi = 1$ near an arbitrary distinct-point configuration. The local representatives agree almost everywhere on overlaps, hence everywhere there by continuity. They glue to S and equal S_0 on the set (7.7). The same argument preserves multilinearity and smearings. [16]

The null-space description in 7.4(ii) follows in both directions: pointwise vanishing implies vanishing smearings, while smoothness and testing in small neighborhoods recover pointwise vanishing from (7.13). The identities (7.15) then hold pointwise. A null input has a null image under κ or δ, since all its correlations, and hence their derivatives, vanish. Thus the maps descend to F_d. Applying the same identities to all other fields proves injectivity of $j_{d,e}$, its composition law and $j_{d,d} = 1$. □

The symmetries and positivity of the Wilson measure survive the ordered limit. The scale condition (7.2) also converts the bounds at fixed cell sizes into bounds in terms of the distance between insertions. We spell these out because they are the inputs to the later Hilbert-space construction.

For a coordinate reflection r_μ, a positive configuration has $x_i^\mu > 0$ and pairwise distinct points. For two such lists of lengths n and m, define the reflected correlation

$$G_S((v,x),(w,y)) = S_{n+m}(\rho(r_\mu)\bar{v}, w; r_\mu x, y). \tag{7.17}$$

Reflection acts on every point and coefficient in the first list. For joint separated tests f_α of Definition 2.10 supported on positive configurations, put

$$G_{\alpha\beta}(f) = \iint \overline{f_\alpha(x)} f_\beta(y) G_S((v^\alpha,x),(v^\beta,y))\,dx\,dy. \tag{7.18}$$

These are the Osterwalder–Schrader forms for point insertions and their smearings. In the estimate below, D_x^m is the mth Fréchet derivative, equipped with its operator norm.

Proposition 7.5 (Euclidean properties)**.** *The kernels of Proposition 7.4 satisfy the following identities and bounds.*

(i) *For the coefficient action and units of Proposition 6.7, every permutation σ, $h \in H_4$, $b \in \mathbb{R}^4$ and distinct configuration x of (2.10),*

$$\begin{aligned} S_\varnothing &= 1, & S_{n+1}(u_d, v; x_0, x) &= S_n(v; x),\\ S_n(v\circ\sigma; x\circ\sigma) &= S_n(v;x), & S_n(\bar{v};x) &= \overline{S_n(v;x)},\\ S_n(\rho(h)v; hx+b) &= S_n(v;x). \end{aligned} \tag{7.19}$$

(ii) *For every coordinate direction and finite collection of positive configurations, the matrices in (7.17)–(7.18) satisfy*

$$[G_S((v^\alpha,x^\alpha),(v^\beta,x^\beta))]_{\alpha,\beta} \succeq 0, \qquad [G_{\alpha\beta}(f)]_{\alpha,\beta} \succeq 0. \tag{7.20}$$

[16] Formal correspondence at revision `3456e5e`: `Dash/Regular.lean`, `Dash/Twist.lean`, and `Org/Input.lean`: smooth representatives with the retained extraction.

(iii) *For every $n, m \in \mathbb{N}$, degree list (d_i) and coefficient list $v_i \in V_{d_i}$, there are $C \geq 0$ and $b \in \mathbb{N}$ such that*

$$\|D_x^m S_n(v;x)\| \leq C r^{-b} \quad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,. \tag{7.21}$$

The identities descend through π_d to the quotient fields.

Proof. Exact lattice symmetries and unit identities pass at grid configurations in (7.7) and grid translations. Density and continuity give these identities throughout Conf_n and for every real translation. For reflection positivity, fix finitely many joint tests supported in positive-side separated configurations. Their smeared lattice observables are strictly inside the positive half-torus for large p and then large N, since each insertion depends on finitely many links whose physical distances from its anchor tend to zero. Lemma 7.1 makes their Gram matrix positive semidefinite. Every entry is a separated smear: its two halves are internally separated and lie strictly on opposite sides of the plane. Pass each entry to its ordered limit. The limiting integrals converge absolutely by smoothness on their compact supports, and the positive-semidefinite cone is closed. Normalized smooth bumps concentrating at fixed positive configurations give point reflection positivity.

For the position-independent bound, (7.2) implies $a_{j+1} \geq c a_j^2$ for some $c > 0$, after adjusting finitely many levels. The first $j \geq j_0$ with $a_j < \delta \leq 1$ therefore satisfies $a_j \geq c_{j_0} \delta^2$. For $n \geq 1$ and fixed v, set $D = 1 + \max_i d_i$. Choose the first sufficiently large m with $a_m < \frac{r}{2}$, then the first $k \geq m$ with $a_k < \frac{a_m}{6(n+1)}$. The two first-crossing bounds give $a_k \geq c' r^4$. Equation (7.4) with $\varepsilon = 1$ and the separated-cell argument yield

$$|S(v;x)| \leq C r^{-4nD} \qquad (0 < r \leq 1,\ \forall i \neq j,\ |x_i - x_j|_\infty \geq r)\,.$$

First pass the estimate at points of (7.7), then use density, retaining separation $\frac{r}{2}$ before taking limits. Repeated derivative identities apply this bound to finitely many lists of higher-degree coefficients. A finite-dimensional norm comparison then bounds each Fréchet derivative norm. Arity zero uses $S_\varnothing = 1$. [17] □

The null-space characterization 7.4(ii) also gives the field symmetries. Inclusions commute with conjugation and H_4 and preserve unit classes. Derivatives commute with conjugation; the chain rule in $S(\rho(h)v; hx) = S(v;x)$ gives

$$\rho_{d+1}(h)\partial_\mu = \sum_\nu h_{\nu\mu}\partial_\nu \rho_d(h) \qquad (h \in H_4)\,.$$

Continuous rotation covariance will require a further argument. [18]

7.5 Removing the spacing refinement

Use the full fixed-volume convergence of Proposition 5.1. For a test supported inside volume p in Definition 2.11, put

$$\widehat{R}_p(v;f) = \lim_{N\to\infty} C_{N,p}(v;f)\,.$$

Restriction to N_j preserves this limit. Uniqueness in $\mathbb{C}$ and (7.9) give $\widehat{R}_p(v;f) = R_p(v;f)$ at every volume containing the test support. The outer limit and smooth correlation functions are retained.

[17] Formal correspondence at revision `3456e5e`: `Extraction/Harvest.lean`: symmetry and positivity; `Work/D05.lean`: two-scale bound; `Work/B10.lean`: derivative bounds.

[18] Formal correspondence at revision `3456e5e`: `Work/D06.lean`: orientation of the derivative maps.

Start the extraction with the full spacing sequence and the prescribed volume sequence p, and compose all the volume refinements into one strictly increasing q. Then

$$\lim_{k\to\infty} \widehat{R}_{p(q(k))}(v; f) = \int f(x) S(v; x)\, dx\,. \tag{7.22}$$

At volumes not containing the test support, the numbers may be assigned arbitrarily, as in Definition 2.12. This is precisely the ordered Wilson limit of Definition 2.12, for every coefficient list and separated test of Definition 2.10. [19]

Unresolved. The full-spacing fixed-volume theorem for the chosen Wilson family remains to be proved in the ultraviolet part, using its fixed-level Cauchy estimates and positional-limit construction. The uniqueness argument above uses this input; it does not prove it.[20]

The smooth null-space definition agrees with the one using separated tests. Consequently the projection $\pi_d(v) = [v]$ of Proposition 7.4 identifies lattice representatives with quotient fields in every separated correlation. Multiply that equality by f to transport (7.22) to the field correlations in Proposition 5.2. The remaining phase assertions of that proposition require the later construction on these same correlations.

References

[1] OpenAI, *Lean construction of four-dimensional Yang–Mills theory*, Source release, revision `3456e5e`, 2026, `https://github.com/yang-mills-group/OpenAI-YM-20261028`.

[19] Formal correspondence at revision `3456e5e`: `FullVolume/Finale/Program.lean`: removal of the spacing refinement; `HierarchyBridge.lean`: public ordered Wilson limit.

[20] Formal correspondence at revision `3456e5e`: `FullVolume/Fields/Filled.lean`: full-spacing fixed-volume convergence for the chosen run.